Wireless Access Electromagnetics

Report Lab 1

Sheharyar Ahmed Nizamani
S329580

Exercise 1:

Parameters case 1(ZL=100)

Design frequency= 5.8 GHz,

W= 1.65 mm

Eeff=3.33

L=7.09mm

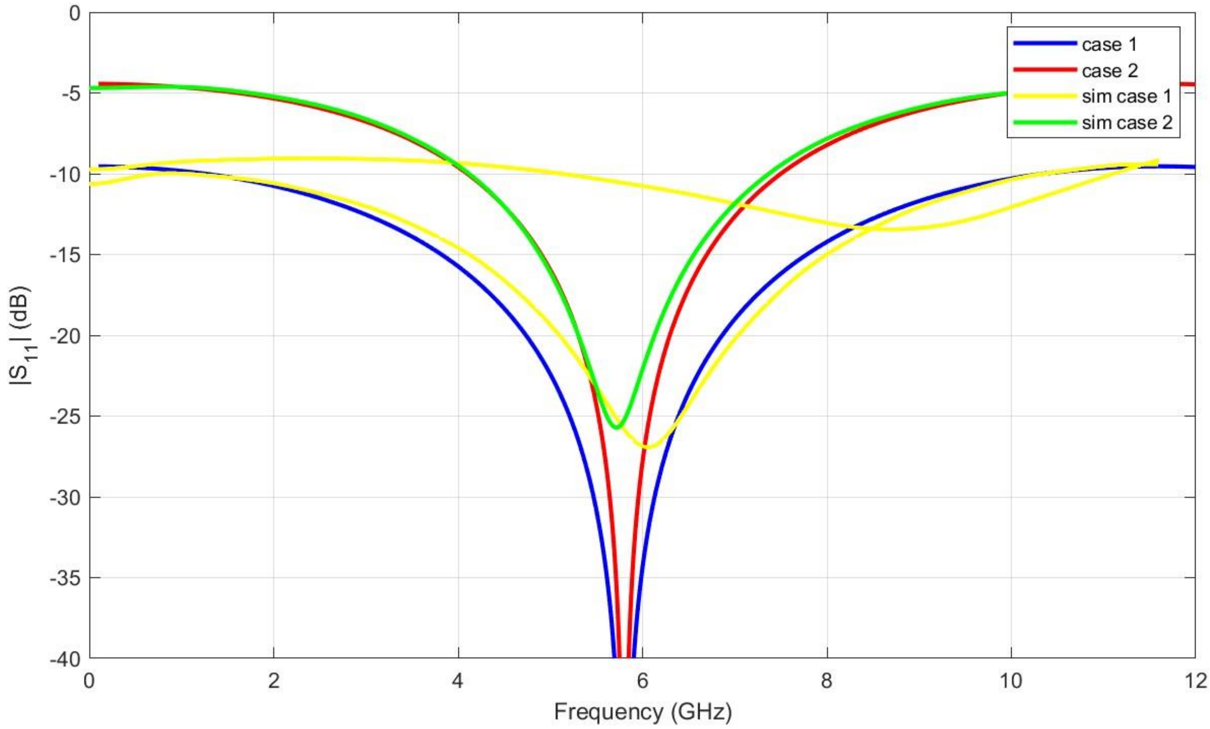
Parameters case 2(ZL=200)

Frequency= 5.8 GHz

W=0.68

Eeff= 3.12mm

L= 7.31mm



Comments:

In Exercise 1, my CST simulations (**sim case 1 & 2**) closely match the theoretical **5.8 GHz** resonance, confirming that the quarter-wave sections were sized correctly for the FR4 substrate, although the finite dip of -25 dB reflects the real-world dielectric and conductor losses ignored by the MATLAB calculations. The wider matching bandwidth of **Case 1** compared to **Case 2** confirms the principle that lower impedance transformation ratios are less frequency-sensitive

Exercise 2:

Parameters:

Case 1:(100ohms)

Frequency=5.8 GHz

W= 0.65 mm

Eeff= 3.33

L = 14.1 mm

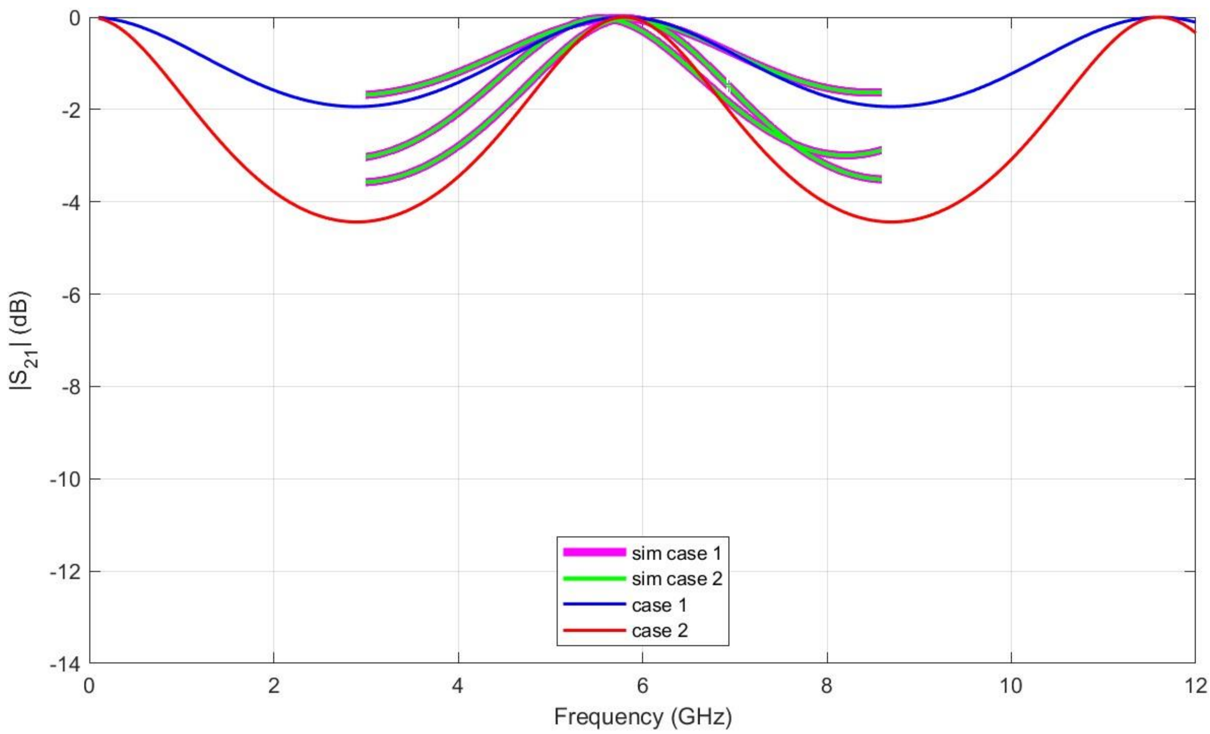
Case 2:(150ohms)

Frequency=5.8 GHz

W= 0.174 mm

Eeff= 3.12

L = 15.2 mm



Comments:

Both simulation cases hit a transmission peak (S21 approx 0 dB) at **5.8 GHz**. I noticed that the peak for **Case 2** is significantly sharper than **Case 1**, proving that a higher impedance contrast with the 50ohm ports increases the quality factor and narrows the bandwidth. My CST results are a bit below the ideal 0 dB theoretical line because the simulation also accounts for the actual dielectric and copper losses of the FR4 substrate.